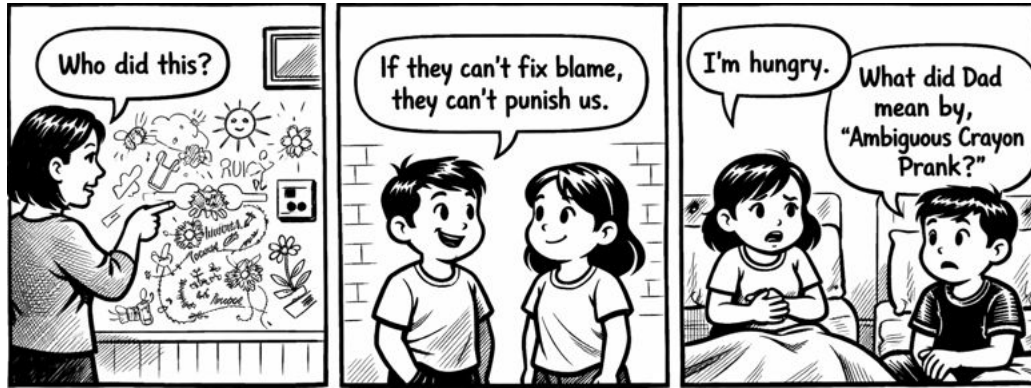


Chapter 17

“The Ambiguous Collapse Paradigm”



Credit – ChatGPT via author guidance: “Crayon Masterpiece”

Consider the industries’ canonical example of two spacelike separated observers, Alice and Bob, who share a stream of EPR style entangled photons, say in the anti-symmetric state.

$$\Psi_{12} = \frac{1}{\sqrt{2}} \{ |H\rangle_1 |V\rangle_2 - |V\rangle_1 |H\rangle_2 \} \quad (1)$$

The two photons do not have independent states, they are in a non-separable state, an entangled state. In the orthodox description of quantum mechanics, a measurement on one *instantaneously* collapses this entangled state into a pair of pure states (relativity be damned), either

$$\begin{aligned} \Psi_1 &= |H\rangle_1 \\ \Psi_2 &= |V\rangle_2 \end{aligned} \quad (2)$$

or

$$\begin{aligned} \Psi_1 &= |V\rangle_1 \\ \Psi_2 &= |H\rangle_2 \end{aligned} \quad (3)$$

Now here is the rub, either observer is free to *claim* that they receive a photon in the entangled state (Equation 1), that it is their measurement which causes collapse to a pair of pure states for each photon (equations 2 & 3) and that the other observer merely receives a stream of photons in a random mixture of states in whatever basis the *first* observer chose.

Either observer believes it is their observation which causes wave function collapse. These two claims are not compatible with each other, the wave function is regarded as a real physical thing, not just math, not just bookkeeping, not some measure of ignorance, but that it is a true element of reality.

To see how incompatible these two stories of causality are, let them make their measurements in conjugate bases. Alice makes her measurements in the *orthogonal* basis (horizontal/vertical) while Bob makes his measurements in the *angled* basis (diagonal/slant).

Alice's Story

Alice believes she is receiving a series of entangled photons

$$\Psi_{12} = \frac{1}{\sqrt{2}} \{ |H\rangle_1 |V\rangle_2 - |V\rangle_1 |H\rangle_2 \} \quad (4)$$

that she is measuring *first*, thus, Bob is receiving a random mixture of pure states in the orthogonal basis

$$\begin{aligned} \Psi_2 &= |V\rangle_2 \\ \Psi_2 &= |H\rangle_2 \end{aligned} \quad (5)$$

She claims that at some specific spacetime location his photons spontaneously change from an entangled state to a pure state.

But because Bob is going to measure in the angled basis, he is actually receiving photons in a superposition of states,

$$\Psi_2 = \frac{1}{\sqrt{2}} \{ |D\rangle_2 \pm |S\rangle_2 \} \quad (6)$$

which upon his measurement collapse a second time from the orthogonal basis to the angled basis.

$$\begin{aligned} \Psi_2 &= |D\rangle_2 \\ \Psi_2 &= |S\rangle_2 \end{aligned} \quad (7)$$

She believes her photons change state once, but his change states twice.

Bob's Story

In perfect symmetry, Bob believes he is receiving a series of entangled photons

$$\Psi_{12} = \frac{1}{\sqrt{2}} \{ |D\rangle_1 |S\rangle_2 - |S\rangle_1 |D\rangle_2 \} \quad (8)$$

that he is measuring *first*, thus, Alice is receiving a random mixture of pure states in the angled basis

$$\begin{aligned} \Psi_1 &= |V\rangle_1 \\ \Psi_1 &= |H\rangle_1 \end{aligned} \quad (9)$$

He claims that at some specific spacetime location her photons spontaneously change from an entangled state to a pure state.

But because Alice is going to measure in the orthogonal basis, she is actually receiving photons in a superposition of states,

$$\Psi_1 = \frac{1}{\sqrt{2}}\{|H\rangle_1 \pm |V\rangle_1\} \quad (10)$$

which upon her measurement collapse a second time from the angled basis to the orthogonal basis.

$$\begin{aligned} \Psi_2 &= |D\rangle_2 \\ \Psi_2 &= |S\rangle_2 \end{aligned} \quad (11)$$

He believes his photons change state once, but hers change states twice.

ACP

These two causal stories assign different states to the photons along the exact same worldlines, and different points along those worldlines when those states change. They describe different histories, incompatible histories, a different chain of causality. The conundrum? Both stories predict that Alice will see a random sequence of horizontal and vertical polarizations and Bob will see a random sequence of diagonal and slant polarizations. Their predictions are story agnostic.

SSI

The attractiveness of the symmetric spacetime interval was that it's zipline allowed the two observers to agree on the story, to claim that this causal chain was the one true chain of causation, the one true physics. It represented progress at defining a real spacelike cause.

QTP offered a way around temporal paradox and thus offered hope that the universe could be made safe for spacelike causes. It pretty much demanded a single causal story, and it might still provide that, but now we have evidence that dual stories are not devastating. Even Einstein might have preferred the single story version of physics.

But now we are not sure. Now we have to keep both possibilities in mind – nature can still surprise us.

Royal Pawns

What is disturbing about the royal pawns position is that we had an inherent ambiguity, a latent superposition, and we found that so intolerable, that without even thinking about it, we enforced a classical simplification, invented a notation which mandated one-and-only-one game. Evolution has programmed a very strong algorithm into us, we don't *want* the Universe to be quantum.

But for a moment, just a moment, we actually had a quantum system, a non-classical toy universe, right under our very noses, and soundly rejected it. Paradigms are pernicious.